

Rot Clock: Team Brief

Climate Hack-tion 2026 · Prepared 3 Oct 2026

Rot Clock is a game-style fridge app that ranks food by how soon it spoils and shows the money and methane saved when you eat it in time. We must submit on Junction by 9:00 pm Sun 4 Oct, Sydney time (AEDT).

At a glance

- **Project name:** Rot Clock
- **One-line pitch:** Save food. Save money. Save the climate. One tap does all three.
- **Challenge:** Build for 2035, track Zero Waste and Methane Reduction. It also supports Awareness across all areas.
- **Target users:** students in share-houses and apartments, and households in general. Later: campuses and councils.
- **Built so far:** a clickable web prototype with four tabs (Fridge, Add, Cook, Impact).
- **Prototype link:** <https://claude.ai/artifact/Dd6gDQVTW3meVZGBUNRv2S> (private until the owner shares it)
- **Team nationality for submission:** to confirm (Australia, New Zealand or Pacific Islands).

Problem statement

People waste food they already paid for, and that waste becomes methane in landfill, yet most tools only help after the waste exists.

- We buy food, forget it, and it goes bad.
- We throw it away, so the money is wasted too.
- Food in landfill rots without oxygen and makes methane, a gas that warms the planet.
- Most tools help us sort waste after it exists. Few help us stop it happening.
- People can't see how small daily choices affect their money and the climate, so habits don't change.
- Open item: add one sourced statistic here (household food waste, or organic waste and methane) with its link.

Solution

Rot Clock turns your fridge into a countdown and a game, so using food up in time pays you back in points, money, methane and a meal.

1. **Add food.** Type it or tap a quick-add button, with an optional price. The app estimates a rot date from typical shelf-life guidance.
2. **See the countdown.** Red means eat soon, amber means 2 to 3 days, green means fine for now.

3. **Get a rescue meal.** For food about to spoil, the app suggests what to cook, and builds a 3-day plan around what you already have.
4. **Play.** Eating food in time earns points, streaks and levels, and keeps a fridge garden healthy. Waste food and it wilts.
5. **Track results.** Every save updates food saved, money saved and estimated methane avoided.
6. **Shop smarter.** A check before you buy warns you if you already have the item.

The core idea: one tap on "Ate it" gives four rewards at once (points, savings, climate impact, a meal).

Prototype

The prototype is a working single-page web app, built in one file, that runs in any browser. Open the [live prototype](https://claude.ai/artifact/Dd6gDQVTW3meVZGBUNRv2S) at <https://claude.ai/artifact/Dd6gDQVTW3meVZGBUNRv2S> (it is private until the owner shares it). For the public repo, use the `index.html` file shared with the team.

Tab	What it does
Fridge	Fridge garden, level, points, streak and a save-5-items challenge. Totals for food, money and methane. An "Eat first" list with a countdown ring per item and Ate it / Binned it buttons.
Add	Type-ahead search, one-tap quick add for 12 common foods, and custom items.
Cook	Rescue recipes ranked by what expires first, a 3-day plan, a vegetarian or vegan filter, and a "before you shop" check.
Impact	Methane avoided vs released, where the grocery money went, an "if more households did this" slider, and the assumptions panel.

Demo helpers: the Next day button in the header moves the clock forward so expiry and the wilting garden show live. Load sample data and Start empty are on the Impact tab.

Not built yet (say so in the pitch): receipt scanning, barcode scanning, leaderboards, AI-generated recipes. Recipes today are fixed templates.

Impact and COP31 alignment

Rot Clock cuts waste at the source, which is the most direct way to reduce landfill methane and meets COP31's goal of slowing waste growth by 50% by 2035.

- **For users:** less food wasted, lower grocery bills, and a visible link between daily habits and climate impact.
- **For the climate:** less organic waste reaches landfill, so less methane. All figures are estimates with assumptions shown in the app.
- **At scale:** campuses and councils could use anonymised data to see where waste happens and target education.
- **Priority fit:** Zero Waste and Methane Reduction (main). Awareness across all areas (secondary), because the app makes climate impact understandable in everyday terms.

Numbers to verify before submitting

Every number in the prototype is an estimate, and the methane figures were written from memory, so someone on the team must check each source below and cite it in the pitch.

The methane estimate for landfilled food waste uses IPCC 2006 default factors:

$$1000 \text{ g} \times 0.15 \times 0.5 \times 1.0 \times 0.5 \times 16/12 = 50 \text{ g CH}_4 \text{ per kg}$$

Item	Value in prototype	What to do
Carbon share of food waste	0.15	Check against the IPCC 2006 Guidelines, Vol. 5, Ch. 3, and cite
Share that decomposes	0.5	Same source
Landfill type factor	1.0	Same source
Methane share of landfill gas	0.5	Same source
Warming factor for methane	28 (100 years)	Check against IPCC AR5 and cite
Landfill gas capture	0% by default	Find a figure for your region, or keep it as an adjustable setting
Shelf lives, pack weights, prices	Placeholder values	Check against food-safety guidance and local prices

In the pitch, say "estimated methane avoided", and never present shelf-life dates as food-safety guarantees.

How we score against the judging criteria

COP31 alignment and build quality are 30% each, so the demo must state the priority clearly and show the core loop working. Ties go to the higher COP31 alignment score.

Criterion	Weight	What we show
COP31 alignment	30%	Name the priority and the 2035 goal on the first slide. Explain why food waste matters for methane.
Build quality	30%	Live loop: add food, countdown, rescue meal, "Ate it", and the totals update. Show the assumptions panel.
Creativity	20%	Prevention instead of sorting, a fridge garden that wilts, and one tap that rewards points, money and climate.
Presentation clarity	20%	One user (a student in a share-house), one problem, one action. Plain language throughout.

Roles and plan

We have until 9:00 pm Sun 4 Oct (Sydney time), so we aim to submit by about 6:00 pm and keep three hours as a buffer. Fill in the owner column.

Role	Responsibility	Owner
Build lead	Repo, GitHub Pages link, fixes to the prototype	
Research and numbers	Verify every source in the numbers section, write the statistic for the problem statement	
Pitch writer	Junction text fields, README, tools and AI disclosure	
Demo lead	Record and edit the 2-minute video	
Submitter	Owns the Junction account, submits and checks every link opens without a login	

1. **Now:** agree owners, create a public GitHub repo, add `index.html` and a README, turn on GitHub Pages.
2. **Tonight:** research owner verifies sources. Build lead fixes anything found while testing on a phone.
3. **Sun morning:** pitch writer drafts all Junction fields. Demo lead rehearses the script.
4. **Sun midday:** record the video (2 minutes or less, made this weekend). Re-record if needed.
5. **Sun about 3:00 pm:** fill in Junction with everything except the final link checks. Save progressively.
6. **Sun about 6:00 pm:** open every link in a private window, confirm the submission shows as complete, and stop building.
7. **Sun 9:00 pm:** hard deadline. Only small bug fixes are allowed after time is up.

Submission checklist

Submit on Junction progressively, because uploads can fail near the deadline, and tick each item as it is done.

- ☐ Project name and one-sentence summary
- ☐ Challenge selected (Zero Waste and Methane Reduction)
- ☐ Team nationality (Australia, New Zealand or Pacific Islands, by where most of the team is from)
- ☐ Team members and roles listed correctly (3 to 5 people)
- ☐ Problem and target users
- ☐ Solution description and intended impact
- ☐ Written pitch: problem, solution and approach
- ☐ Demo video, 2 minutes or less, made this weekend
- ☐ Public repository link, kept live until winners are announced
- ☐ Supporting material, if any
- ☐ Tools used: every outside tool, library, dataset and API, including paid ones
- ☐ AI tools used: list the AI assistant(s) used for code, text and images

- ☐ Disclosure: no prior work, everything was built at the event
- ☐ Every link opens without asking for access
- ☐ Submission is saved and visibly marked complete before 9:00 pm Sun 4 Oct (AEDT)

Two-minute demo script

The video must run 2 minutes or less and focus on the problem, the working solution and evidence. Start from the sample data so the screen is never empty.

1. **Problem (0:00 to 0:15).** "We buy food and forget it. It goes bad, we bin it, and in landfill it makes methane. COP31 aims to slow waste growth by 50% by 2035."
2. **Add food (0:15 to 0:35).** On the Add tab, tap two quick-add foods and type one more. "One tap, and the clock starts."
3. **Countdown (0:35 to 0:55).** On the Fridge tab, point at the red items and the countdown rings. Press Next day twice and show the garden wilt a little.
4. **Rescue meal (0:55 to 1:20).** Open the rescue meal, then press Cooked it. Show the toast with points, food saved and methane avoided.
5. **Impact (1:20 to 1:45).** On the Impact tab show the methane bars, the money breakdown and the assumptions panel. "Every number shows its source and assumptions."
6. **Scale and close (1:45 to 2:00).** Drag the households slider. "Save food. Save money. Save the climate. One tap does all three." Mention what comes next: receipt scanning and campus dashboards.

Record on a phone-width window if you can, and do one dry run first.

Rules to remember and contacts

Breaking the hacking rules can lead to disqualification, and organisers can ask to see our code, so we follow these and ask early when unsure.

- Build from scratch during the event. No pre-built code, designs or assets.
- AI tools, open-source libraries, datasets and APIs are allowed, but every one must be disclosed in the submission.
- The repository must be public and stay live until winners are announced.
- Stop building at the deadline. Small bug fixes only after that.
- Everyone on the team takes an active part.
- Closing ceremony is Mon 12 Oct. The guide gives two different times (3:00 pm and 4:00 pm AEDT), and the Webex link is still to be confirmed, so check Discord.

Need	Contact
General questions	Discord #general
Technical help	Discord #getting-started
Rules questions	Discord #faq-and-resources
Genuine technical failure near the deadline	Email shelly.amir@gdsi.ie before the deadline, with timestamped evidence
Conduct or safety	shelly.amir@gdsi.ie or sharon.offenberger@gdsi.ie
Event hub	hackjunction.app/hackathons/climate-hack-tion